SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5 **Duration : 1 Hour**
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Annexure A — Analytical Problem Solving

1. Water Jug Problem

Solution (4-gallon jug and 3-gallon jug):

1. Fill the 3-gallon jug and pour it into the 4-gallon jug. (4-gallon jug: 3, 3-gallon jug: 0)
2. Fill the 3-gallon jug again and pour into the 4-gallon jug until it is full. Only 2 gallons can fit, so 1 gallon remains in the 3-gallon jug.
3. Empty the 4-gallon jug onto the ground.
4. Pour the remaining 1 gallon from the 3-gallon jug into the 4-gallon jug.
5. Fill the 3-gallon jug and pour it into the 4-gallon jug. The 4-gallon jug now contains exactly **2 gallons**.

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Annexure A — Analytical Problem Solving

2. Mars Rover as an Intelligent Agent

i. Percepts: Camera images, terrain and obstacle data, distance sensors, wheel position, temperature, and scientific instrument readings.
ii. Operating environment: Mars terrain; partially observable, uncertain, sequential, mostly static during a short action, and physically continuous.
iii. Actions: Move or turn, stop, take images, collect/analyse samples, operate instruments, and send data to Earth.
iv. Performance measure: Safe navigation, successful sample analysis, useful scientific data, task completion, and efficient use of time and power.
v. Suitable architecture: A **model-based, goal-based agent**. It maintains an internal map/state because the environment is only partly observable and plans actions to reach scientific goals while avoiding hazards.

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Annexure A — Analytical Problem Solving

3. 8-Queens Problem Formulation

Goal: Place 8 queens on an 8 × 8 chessboard so that no two queens attack each other.
State: A partial arrangement of queens, normally placing one queen in each of the first few columns.
Initial state: An empty chessboard.
Actions: Place a queen in a safe square in the next column.
Transition model: The selected square gains a queen, creating a new board configuration.
Goal test: Eight queens are placed and no pair shares a row, column, or diagonal.
Path cost: Usually 1 for each queen placed; all complete solutions therefore cost 8.
Search strategy: Backtracking/depth-first search can place queens column by column. If a placement causes a conflict, undo it and try another safe square.

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Annexure A — Analytical Problem Solving

4. OLA Cab Booking as a Problem-Solving Agent

Agent type: A goal-based problem-solving agent, with cost-based route selection if the user wants to minimise fare or travel time.
Initial state: Customer's current location.
Goal state: Customer reaches the selected destination.
Actions: Select cab type (Mini, Micro, Sedan, Shared, Prime, etc.), choose a route, book the cab, and travel.
Evaluation: Destination reached while satisfying the user's preference for fare, time, comfort, and availability.
Pseudocode:

```
START
Read pickup, destination, and preferences
Get available cab types and route options
Remove options that do not meet preferences
Choose the best option by cost, time, and comfort
Show fare and estimated arrival time
IF customer confirms THEN book cab
Track trip until destination is reached
END
```

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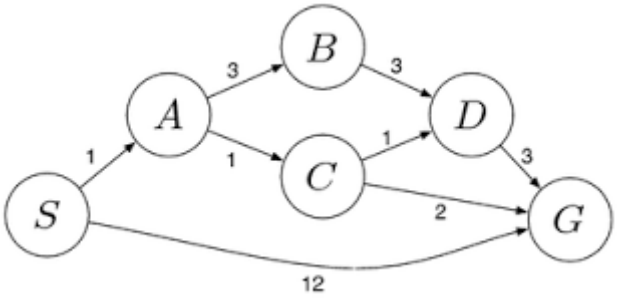
Annexure A — Analytical Problem Solving

5. Uniform Cost Search (UCS)

Purpose: UCS finds a least-total-cost path from start node S to goal node G when edge costs are non-negative.
Method: Expand the frontier node with the smallest accumulated path cost, not the node with the fewest edges.
Pseudocode:

```
UCS(start, goal):
  frontier = priority queue containing (0, start)
  cost[start] = 0; parent[start] = None
  while frontier is not empty:
    (g, node) = remove lowest-cost entry
    if node == goal: return path(parent), g
    for each neighbour v with edge cost w:
      new_cost = g + w
      if v is unseen or new_cost < cost[v]:
        cost[v] = new_cost; parent[v] = node
        insert (new_cost, v) into frontier
  return failure
```

Note: The uploaded PDF mentions a weighted graph, but the graph itself is not visible in the provided pages. Therefore, the exact S-to-G path and total cost cannot be calculated from this file alone.



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

		requirements accurately			